1. a) Adder :

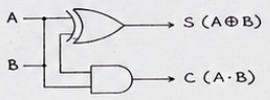
Aaux : An adder is a combinational circuit that performs arithmetic addition of binary numbers. It is widely used in ALU, counters, microprocessors etc.

- **Half Adder :** Adds two 1-bit binary numbers. It has two outputs : Sum (S) and Carry (C).

Truth Table :

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Logic Diagram :

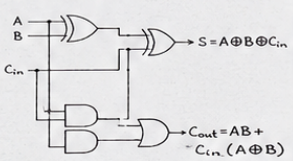


- **Full Adder :** Adds three 1-bit inputs (A, B and Carry in (C_{in})). Outputs are Sum (S) and Carry out (C_{out}).

Truth Table :

A	B	C_{in}	S	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Logic Diagram :



Boolean Expressions :

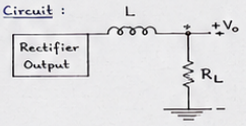
$S = A \oplus B \oplus C_{in}$

$C_{out} = AB + C_{in}(A \oplus B)$

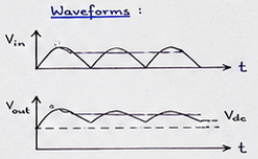
b) L-filter (Inductor filter) :

An inductor connected in series with the load. It offers high reactance to AC ripple and low resistance to DC. Used for high current applications.

Circuit :



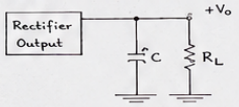
Waveforms :



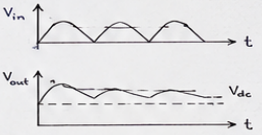
C-filter (Capacitor filter) :

A capacitor is connected in parallel with the load. It offers low reactance path to ripple and hence removes AC components. Used for low current applications.

Circuit :

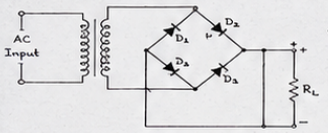


Waveforms :

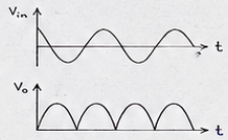


2. a) Full Wave Rectifier :

A full wave rectifier utilizes both half cycles of AC input. It can be implemented using two diodes with center tapped transformer or four diodes bridge rectifier. ° (Bridge Rectifier circuit)



Waveforms :



Average Output Voltage (V_{av}) :

$$V_{av} = \frac{2V_m}{\pi}$$

where V_m = peak value of secondary voltage.

Average Output Current (I_{av}) :

$$I_{av} = \frac{2I_m}{\pi} = \frac{V_{av}}{R_L}$$

b) Encode 1100110101 into odd parity Hamming code.

Given data bits = 1100110101 (10 bits)

Number of parity bits r such that $2^r \geq m + r + 1$

Here m = 10

$2^4 = 16 \geq 10 + 4 + 1 = 15$ so, r = 4

Total bits = 10 + 4 = 14

Parity bit positions : 1, 2, 4, 8

Place data bits in remaining positions.

Positions :	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Bits :	$\overline{P_1}$	$\overline{P_2}$	1	$\overline{P_4}$	1	1	0	$\overline{P_8}$	0	1	1	0	1	0

Now calculate parity bits for odd parity.

P_1 covers positions : 1, 3, 5, 7, 9, 11, 13
→ 1,1,0,0,0,1 → total 3 ones (odd) → P_1 = 0

P_2 covers positions : 2, 3, 6, 7, 10, 11, 14
→ 1,1,0,1,1,0 → total 4 ones (even) → P_2 = 1

P_4 covers positions : 4, 5, 6, 7, 12, 13, 14
→ 1,1,0,0,1,0 → total 2 ones (even) → P_4 = 1

P_8 covers positions : 8, 9, 10, 11, 12, 13, 14
→ 0,0,1,0,1,0 → total 3 ones (odd) → P_8 = 0

Final Hamming Code (14 bits) =

0	1	1	1	1	1	0	0	0	1	1	0	1	0

3. a) DC Regulated Power Supply :

It converts AC mains into constant DC voltage using rectifier, filter, regulator and other components.



Explanation :

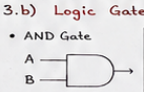
- Transformer steps down the AC voltage.
- Rectifier converts AC to pulsating DC.
- Filter removes ripples and provides smooth DC.
- Regulator maintains constant DC voltage even if input or load varies.

Advantages : Good regulation, low ripple, safe & reliable.

Applications : Used in electronic circuits, communication devices, computers, instrumentation etc.

3.b) Logic Gates with Truth Tables :

- **AND Gate**



A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

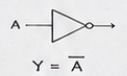
$Y = A \cdot B$

- **OR Gate**

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

$Y = A + B$

- **NOT Gate**



A	Y
0	1
1	0

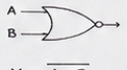
$Y = \overline{A}$

- **NAND Gate**

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

$Y = \overline{A \cdot B}$

- **NOR Gate**



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

$Y = \overline{A + B}$

- **XOR Gate**

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

$Y = A \oplus B$

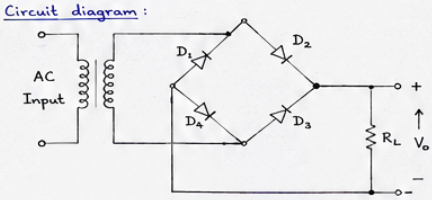
More Matter / Extra Points :

1. Half Subtractor : $D = A \oplus B$, Borrow = $\overline{A}B$
2. Full Subtractor : $D = A \oplus B \oplus Bin$, Borrow = $\overline{A}B + \overline{A}Bin + BBin$.
3. Ripple Carry Adder : Cascade of full adders where C_{out} of one is C_{in} of next.
4. Applications of Adders : ALU, Counters, Address calculation, Digital Signal Processing.
5. Filters : π-filter (C-L-C) gives better smoothing.
6. Other Rectifiers : Half Wave Rectifier (uses one half cycle only).
7. Regulation Methods : Series regulator, Shunt regulator, IC regulators.
8. More Logic Gates : XNOR (equivalence), Buffer, Tri-state buffer.

4. a) Bridge rectifier :

A bridge rectifier uses four diodes connected in a bridge form. It converts both half cycles of AC into DC. It does not require center tapped transformer.

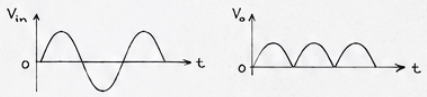
Circuit diagram :



Operation :

- During positive half cycle, D_1 and D_3 conduct. D_2 and D_4 are reverse biased.
- During negative half cycle, D_2 and D_4 conduct. D_1 and D_3 are reverse biased.
- In both cases current through load flows in same direction.

Waveforms :



4. b) Encode the given number into odd parity Hamming code.

Given data bits = 1101100 (7 bits)

Number of parity bits r such that $2^r \geq m+r+1$

$2^r \geq 7+r+1 = 8+r$

For $r = 4$, $2^4 = 16 \geq 11$ (OK)

Total bits = $m + r = 7 + 4 = 11$

Parity bit positions = 1, 2, 4, 8

Data bit positions = 3, 5, 6, 7, 9, 10, 11

Place the data bits in order :

Pos :	1	2	3	4	5	6	7	8	9	10	11
Bits :	P_1	P_2	1	P_4	1	0	1	P_8	1	1	0

(from 1101100)

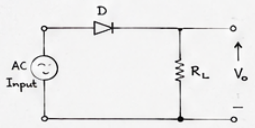
Calculate parity bits for odd parity :

- P_1 covers positions 1,3,5,7,9,11 → 1,1,1,1,0 → 4 ones (even) To make odd → $P_1 = 1$
- P_2 covers positions 2,3,6,7,10,11 → 1,0,1,1,0 → 3 ones (odd) $P_2 = 0$
- P_4 covers positions 4,5,6,7 → 1,0,1 → 2 ones (even) → $P_4 = 1$
- P_8 covers positions 8,9,10,11 → 1,1,0 → 2 ones (even) → $P_8 = 1$

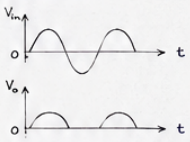
Final Hamming Code (11 bits) :

Pos :	1	2	3	4	5	6	7	8	9	10	11
Bits :	1	0	1	1	1	0	1	1	1	1	0

5. a) i) Half wave rectifier : It uses a single diode to convert only one half cycle of AC into DC.

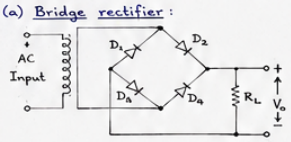


Waveforms :

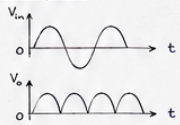


ii) Full wave rectifier : It converts both half cycles of AC into DC. It can be done by (a) Bridge rectifier (b) Centre tapped rectifier.

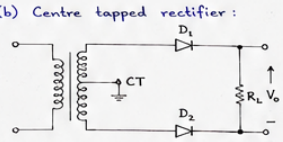
(a) Bridge rectifier :



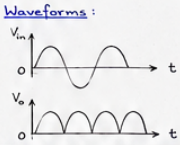
Waveforms :



(b) Centre tapped rectifier :



Waveforms :



5. b) Explain different codes.

a) Hamming code :

It is an error detecting and correcting code. It uses parity bits at positions which are powers of 2 (1, 2, 4, 8, ...). It can detect double bit errors and correct single bit errors.

b) Grey code (Reflected Binary code) :

It is a code in which two successive numbers differ in only one bit. It is used in position encoders, ADC, etc.

Example (3-bit Grey code) :

Decimal :	0	1	2	3	4	5	6	7
Grey :	000	001	011	010	110	111	101	100

c) XS-3 code (Excess-3 code) :

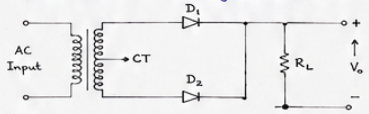
It is a non-weighted code. In this code, 3 is added to the decimal number and then the result is expressed in 8421 BCD. Used in digital systems.

Example :

Decimal :	0	1	2	3	4	5	6	7	8	9
XS-3 :	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101

6. a) Full wave centre tapped rectifier :

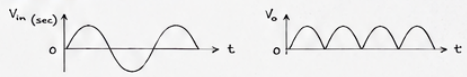
It uses a centre tapped transformer and two diodes. It converts both half cycles into DC.



Operation :

- During positive half cycle, upper end is +ve w.r.t CT. D_1 conducts, D_2 is reverse biased. Current flows through R_L.
- During negative half cycle, lower end is +ve w.r.t CT. D_2 conducts, D_1 is reverse biased. Current through R_L is in same direction.

Waveforms :



6. b) Operation of NAND, NOR, XOR & XNOR gates :

1) NAND Gate :

Output is complement of AND.

$Y = \overline{A \cdot B}$



A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

2) NOR Gate :

Output is complement of OR.

$Y = \overline{A + B}$



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

3) XOR Gate :

Output is 1 when inputs are different.

$Y = A \oplus B = \overline{A}B + A\overline{B}$



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

4) XNOR Gate :

Output is 1 when inputs are same.

$Y = \overline{A \oplus B} = AB + \overline{A}\overline{B}$



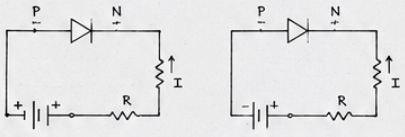
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

UNIT - 4

1. a) P-N junction diode :

A diode is formed by joining p-type and n-type semiconductor. It allows current in one direction (forward bias) and blocks in opposite direction (reverse bias).

Circuit Diagrams :



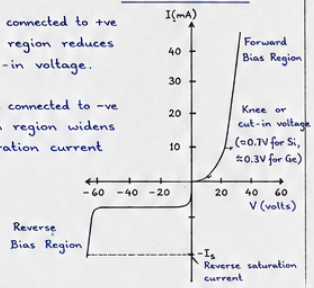
(i) Forward Bias (ii) Reverse Bias

Operation :

(i) Forward Bias – When P side is connected to +ve and N side to –ve, depletion region reduces and current flows after cut-in voltage.

(ii) Reverse Bias – When P side is connected to –ve and N side to +ve, depletion region widens and only small reverse saturation current flows.

V-I Characteristics :



1. b) Transistor biasing :

Biasing is the process of applying suitable DC voltages and currents to the transistor to establish a proper quiescent (Q) point in the active region of its characteristics.

Need for biasing in BJT :

- To operate the transistor in active region for linear amplification.
- To obtain undistorted output signals.
- To make the operation of transistor independent of β (current gain) and temperature variations.
- To provide stability of the operating point (Q-point).
- To ensure maximum power transfer to the load.
- To prevent transistor from entering cut-off or saturation region.

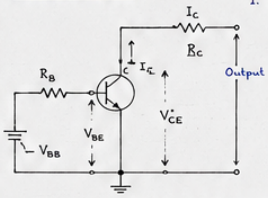
2. a) Comparison between P-N junction diode and Zener diode :

Parameter	P-N Junction Diode	Zener Diode
1. Doping	Lightly doped	Heavily doped
2. Depletion region	Wide	Very thin
3. Reverse breakdown	Occurs at high reverse voltage	Occurs at low reverse voltage (V_z)
4. Operating condition	Normally used in forward bias	Normally used in reverse bias
5. Breakdown mechanism	Avalanche (at high voltage)	Zener effect (low voltage) or Avalanche (high voltage)
6. Applications	Rectification, switching	Voltage regulation, reference voltage

2. b) Common Emitter (CE) mode :

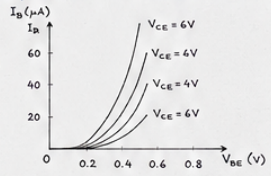
In CE configuration, the emitter is common to both input and output circuits.

Circuit Diagram :

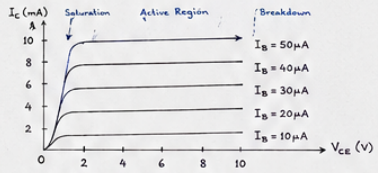


Characteristics :

1. Input Characteristics (at constant V_{CE})



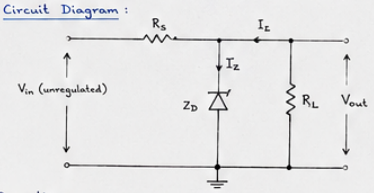
2. Output Characteristics (at constant I_B)



3. a) Zener diode as a Voltage Regulator :

A Zener diode operates in reverse breakdown region. When reverse biased beyond Zener voltage (V_z), it maintains almost constant voltage across it, even if the input voltage or load current changes.

Circuit Diagram :



Operation :

- When V_{in} increases → I_z increases, but $V_{out} \approx V_z$ (constant).
- When V_{in} decreases → I_z decreases, but $V_{out} \approx V_z$.

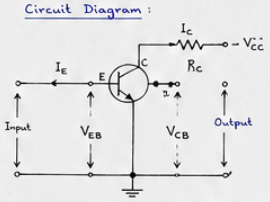
Thus, Zener diode maintains constant output voltage.

Applications : Voltage regulation, reference voltage in power supplies.

3. b) Common Base (CB) mode :

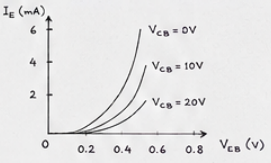
In CB configuration, the base is common to both input and output circuits.

Circuit Diagram :

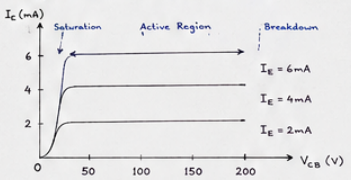


Characteristics :

1. Input Characteristics (at constant V_{CB})



2. Output Characteristics (at constant I_E)



4. a) Define cut-in (knee) voltage and breakdown voltage of a P-N junction diode.

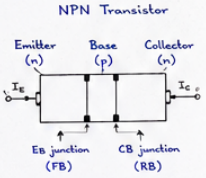
- **Cut-in (knee) voltage (V_γ):**
  The minimum forward bias voltage at which the diode starts conducting significantly and the forward current increases rapidly. After this point, a small increase in voltage causes a large increase in current.

  Typical values:
  - Silicon diode ≈ 0.7 V
  - Germanium diode ≈ 0.3 V

- **Breakdown voltage (V_B):**
  The reverse bias voltage at which the reverse current increases suddenly to a large value with a small increase in voltage, due to breakdown of the p-n junction.

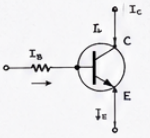
4. b) Describe the operation of BJT (NPN transistor).

An NPN transistor consists of two n-type regions separated by a thin p-type region.



Operation (in active mode):

- The base-emitter junction is forward biased ($V_{BE} \approx 0.7$ V for Si).
- The base-collector junction is reverse biased.
- When a small base current (I_B) flows, electrons are injected from the emitter (n) into the base (p).
- Since the base is very thin and lightly doped, only a few electrons recombine with holes in the base (this small portion equals I_B).
- The remaining electrons diffuse to the collector and are swept into it by the reverse-biased collector junction, producing collector current I_C.
- Hence, a small I_B controls a large I_C.
  ($I_C \approx \beta I_B$, where β is current gain)



5. a) Differentiate between Zener breakdown and Avalanche breakdown.

Parameter	Zener Breakdown	Avalanche Breakdown
1. Mechanism	Due to quantum mechanical tunneling effect.	Due to impact ionization (multiplication of carriers).
2. Doping	Heavily doped junction.	Lightly doped junction.
3. Depletion width	Very thin.	Wide.
4. Breakdown voltage	Low (typically < 5.6 V)	High (typically > 5.6 V)
5. Temperature coefficient	Negative (V_z decreases with temperature).	Positive (V_z increases with temperature).
6. Noise	Low noise.	Higher noise.
7. Electric field	Very high (~10^7 V/m).	Moderate (~10^5 V/m).
8. Occurrence	In diodes designed for low voltage regulation.	In diodes designed for high voltage regulation.

5. b) Compare CE, CB, and CC configurations of a BJT.

Parameter	CE (Common Emitter)	CB (Common Base)	CC (Common Collector)
1. Input terminal	Base	Emitter	Base
2. Output terminal	Collector	Collector	Emitter
3. Current gain (Ai)	High (β)	< 1 (α)	High ($\beta+1$)
4. Voltage gain (Av)	High	High	≈ 1 (slightly less than 1)
5. Power gain	High	Moderate	High
6. Input resistance	Moderate	Low	Very high
7. Output resistance	High	High	Low
8. Phase relation between input and output	180° out of phase	In phase	In phase
9. Application	General purpose amplification	High frequency applications	Buffer/Impedance matching

6. a) Define static resistance and dynamic (AC) resistance of a P-N junction diode.

- **Static resistance (R_s):**
  It is the ratio of DC voltage across the diode to the DC current through it at a given operating point.

$$R_s = \frac{V_D}{I_D} \quad \text{(ohms)}$$

- **Dynamic (or AC resistance (r_d):**
  It is the ratio of the small change in voltage across the diode to the small change in current, at a given operating point in forward bias.

$$r_d = \frac{\Delta V_D}{\Delta I_D} \quad \text{(ohms)}$$

For a forward-biased diode,

$$r_d \approx \frac{\eta V_T}{I_D}$$

where V_T = thermal voltage (≈ 26 mV at 300 K)
η = emission coefficient (≈ 1 for Si diode)

6. b) Explain the difference between NPN and PNP transistors and Explain the operation of NPN transistor.

Difference between NPN and PNP transistors:

Parameter	NPN Transistor	PNP Transistor
1. Construction	n-p-n	p-n-p
2. Majority carriers	Electrons	Holes
3. Conventional current	Enters base ←	Leaves base ←
4. Active mode biasing	BE junction forward biased, CB junction reverse biased.	BE junction forward biased, CB junction reverse biased.
5. Supply polarity	Collector positive w.r.t emitter.	Collector negative w.r.t emitter.
6. Direction of current (Conventional)	C → E	E → C
7. Symbol		

Operation of NPN transistor (active mode):

- The base-emitter junction is forward biased ($V_{BE} \approx 0.7$ V for Si).
- The base-collector junction is reverse biased.
- Electrons are injected from emitter into base.
- Only a small fraction recombines in base (this is I_B).
- The remaining electrons are swept into collector by reverse bias and form collector current I_C.
- Thus, a small base current controls a large collector current.

$I_C \approx \beta I_B$, $I_E = I_C + I_B$

NPN in active mode

